Electricity

30 questions: 10 multiple choice, 20 structured.

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1 · Part 1

Paper 1 & 2 Units [1]

Which row gives the correct unit for each quantity?

	charge	power	resistance
A	ampere	watt	ohm
B	coulomb	watt	ohm
C	coulomb	joule	volt
D	ampere	joule	ohm

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2 · Part 1

Paper 1 & 2 Mains electricity [1]

Which statement about a circuit breaker in a domestic mains circuit is correct?

A It increases the current when there is a fault.

B It is connected in the neutral wire.

C It must be replaced after it has switched off the circuit.

D It can be reset after it has switched off the circuit.

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3 · Part 1

Paper 1 & 2 | Mains electricity | [1]

Which row describes the current supplied by a cell and the current supplied by the mains?

	current supplied by a cell	**current supplied by the mains**
A	direct current (d.c.)	alternating current (a.c.)
B	alternating current (a.c.)	direct current (d.c.)
C	alternating current (a.c.)	alternating current (a.c.)
D	direct current (d.c.)	direct current (d.c.)

4 · Part 1

Paper 1 & 2 | Energy and voltage in circuits | [1]

The circuit contains a light-dependent resistor (LDR) connected in series with a battery and an ammeter. The light shining on the LDR becomes brighter.

Which row shows what happens to the resistance of the LDR and to the ammeter reading?

	resistance of the LDR	**ammeter reading**
A	increases	increases
B	increases	decreases
C	decreases	increases
D	decreases	decreases

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5 · Part 1

Paper 1 & 2 | Energy and voltage in circuits | [1]

What is an electric current in a copper wire?

A. a flow of electrons

B. a flow of protons

C. a flow of positive ions

D. a flow of copper atoms

6 · Part 1

Paper 1 & 2 | Energy and voltage in circuits | [1]

Leila wants to show whether there is a current in a circuit without using a meter.

Which component could she connect in series in the circuit to do this?

A. a fixed resistor

B. a light-dependent resistor (LDR)

C. a thermistor

D. a light-emitting diode (LED)

7 · Part 1

Paper 1 & 2 | Energy and voltage in circuits | [1]

Two lamps, P and Q, are connected in parallel with each other across a 6.0 V battery. The current in lamp P is 0.30 A and the current in lamp Q is 0.20 A.

Which row gives the current from the battery and the voltage across lamp Q?

	current from the battery / A	**voltage across lamp Q / V**
A	0.50	3.0
B	0.50	6.0
C	0.10	6.0
D	0.20	6.0

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8 · Part 1

Paper 1 & 2 | Energy and voltage in circuits | [1]

The table shows how the current in a component changes as the voltage across it is increased.

voltage / V	0	1.0	2.0	3.0	4.0
current / A	0	0.20	0.32	0.40	0.45

Which component could this be?

A a length of wire at constant temperature

B a diode

C a metal filament lamp

D a fixed resistor at constant temperature

9 · Part 1

Paper 2 only | Electric charge | [1]

Which list contains only electrical insulators?

A glass, polythene, rubber

B copper, polythene, rubber

C glass, aluminium, nylon

D nylon, steel, glass

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10 · Part 1

Paper 2 only

Electric charge

[1]

Two small balloons, X and Y, hang from insulating threads. Balloon X is negatively charged. When balloon Y is brought near balloon X, the two balloons push each other apart.

Next, a charged rod Z is held near balloon Y. Balloon Y moves away from rod Z.

Which row shows the charge on balloon Y and the charge on rod Z?

	charge on balloon Y	**charge on rod Z**
A	positive	positive
B	positive	negative
C	negative	negative
D	negative	positive

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11 · Part 1

Paper 1 & 2 | Mains electricity | [6]

The hairdryer in a bedroom is connected to the 230 V mains supply. The power of the hairdryer is 1.8 kW.

(a) State the equation linking power, current and voltage. [1]

(b) Calculate the current in the hairdryer. [3]

(c) The plug of the hairdryer can be fitted with a 3 A, a 5 A or a 13 A fuse.

State which fuse should be fitted. Give a reason for your answer. [2]

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12 · Part 1

Paper 1 & 2 | Mains electricity | [6]

An aquarium heater keeps the water in a fish tank warm. It contains a resistor called a heating element.

(a) Explain why the heating element becomes hot when there is a current in it. [2]

(b) The heater is connected to a 230 V supply. The current in the heater is 0.40 A. The heater is switched on for 2.0 hours.

Use the equation

energy transferred = current × voltage × time

to calculate the energy transferred by the heater, in J. [3]

(c) Give one other domestic appliance that uses the heating effect of an electric current. [1]

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13 · Part 1

Paper 1 & 2 Mains electricity [6]

The washing machine in a house has a metal case. The case is connected to the earth wire. The plug contains a fuse in the live wire.

(a) A fault occurs inside the washing machine. The live wire comes loose and touches the metal case.

Explain how the earth wire and the fuse together protect a person who touches the case. [4]

(b) A hairdryer has a plastic case and is double insulated. It has no earth wire.

Explain why a double-insulated appliance does not need an earth wire. [2]

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14 · Part 1

Paper 1 & 2 Energy and voltage in circuits [5]

(a) State what is meant by an electric current. [1]

(b) State the equation linking charge, current and time. [1]

(c) The current in a phone charger cable is 0.80 A. The phone is charged for 5.0 minutes.

Calculate the charge that flows through the cable in this time. [3]

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15 · Part 1

Paper 1 & 2 | Energy and voltage in circuits | [8]

The battery in a portable radio has a voltage of 9.0 V.

(a) State what is meant by a voltage of 9.0 V. [2]

(b) State the equation linking energy transferred, charge and voltage. [1]

(c) Calculate the energy transferred by the battery when 1500 C of charge flows through the radio. [2]

(d) The current in the radio is 0.050 A.

Calculate the time, in hours, taken for 1500 C of charge to flow through the radio. [3]

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16 · Part 1

Paper 1 & 2 | Energy and voltage in circuits | [6]

The circuit is made from a 9.0 V battery connected in series with an ammeter, a 15 Ω resistor and a 30 Ω resistor.

(a) Calculate the total resistance of the circuit. [1]

(b) State the equation linking voltage, current and resistance. [1]

(c) Calculate the reading on the ammeter. [2]

(d) Calculate the voltage across the 30 Ω resistor. [2]

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17 · Part 1

Paper 1 & 2 | Energy and voltage in circuits | [8]

The circuit is made from a 12 V supply connected in series with an ammeter, a lamp and a fixed resistor. A voltmeter is connected across the lamp.

The voltmeter reads 4.8 V. The ammeter reads 0.30 A.

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(a) Calculate the voltage across the fixed resistor. [1]

(b) Calculate the resistance of the fixed resistor. [3]

(c) Determine the total resistance of the circuit. [2]

(d) The fixed resistor is removed and the lamp is connected directly across the 12 V supply.

State and explain what happens to the current in the lamp. [2]

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18 · Part 1

Paper 1 & 2 | Energy and voltage in circuits | [7]

Sam investigates how the current in a diode varies with the voltage across it.

(a) Describe how Sam could carry out this investigation. Your answer should include the apparatus he needs and how it is connected. [4]

(b) The table shows Sam's results.

voltage / V	**0.0**	**0.2**	**0.4**	**0.5**	**0.6**	**0.7**	**0.8**
current / mA	0	0	0	1	5	18	45

Describe how the current in the diode varies with the voltage across it. [2]

(c) Give a reason why Sam took readings at smaller voltage intervals above 0.4 V. [1]

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19 · Part 1

Paper 1 & 2 | Energy and voltage in circuits | [5]

The ceiling lamps in a house are connected to the mains supply.

(a) Explain why the lamps are connected in parallel rather than in series. [4]

(b) Some decorative lights are made from a string of 20 small lamps connected in series to the mains supply.

Suggest one advantage of connecting these small lamps in series. [1]

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20 · Part 1

Paper 1 & 2 | Energy and voltage in circuits | [8]

Two resistors are connected in parallel with each other across a battery. One resistor has a resistance of 6.0 Ω. The other resistor, R, has an unknown resistance.

An ammeter in series with the 6.0 Ω resistor reads 1.5 A. An ammeter next to the battery, in the main part of the circuit, reads 2.5 A.

(a) Calculate the voltage across the 6.0 Ω resistor. [2]

(b) State the voltage across resistor R. [1]

(c) Calculate the current in resistor R. [1]

(d) Calculate the resistance of R. [2]

(e) Explain why the current is conserved at a junction in a circuit. [2]

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21 · Part 1

Paper 1 & 2 | Energy and voltage in circuits | [5]

The torch contains a 3.0 V battery, a switch and a lamp connected in series.

(a) A second, identical lamp is added in series with the first lamp.

Explain what happens to the current in the circuit. [2]

(b) The 3.0 V battery is then replaced by a 4.5 V battery. Both lamps are still in the circuit.

State and explain what happens to the current in the circuit. [2]

(c) A diode is then added in series in the circuit. It is connected the wrong way round (in reverse).

State the current in the circuit. [1]

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22 · Part 1

Paper 1 & 2 | Energy and voltage in circuits | [8]

The headlamp of a car is connected to a 12 V battery. In 10 minutes, the headlamp transfers 3.6×10^4 J of energy.

(a) Calculate the charge that flows through the headlamp in 10 minutes. [3]

(b) Calculate the current in the headlamp. [3]

(c) Calculate the power of the headlamp. [2]

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23 · Part 1

Paper 1 & 2 | Energy and voltage in circuits | [6]

The temperature sensor in a freezer contains a thermistor. The thermistor is connected in series with a 6.0 V battery and an ammeter.

(a) State how the resistance of a thermistor changes as its temperature increases. [1]

(b) At one temperature, the ammeter reads 12 mA.

Calculate the resistance of the thermistor at this temperature. [3]

(c) The freezer door is left open and the thermistor warms up.

Explain what happens to the ammeter reading. [2]

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24 · Part 1

Paper 2 only

Electric charge

[5]

Mia holds a copper rod in her hand and rubs it with a dry cloth. The copper rod does not become charged. She then holds a polythene rod in her hand and rubs it in the same way. The polythene rod becomes charged.

(a) Explain these observations in terms of the movement of electrons. [3]

(b) Mia fixes a plastic handle to the copper rod. She holds only the plastic handle and rubs the copper rod again.

Suggest whether the copper rod becomes charged. Give a reason for your answer. [2]

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25 · Part 1

Paper 2 only

Electric charge

[7]

Omar investigates how insulating materials can be charged by friction.

One polythene rod, which is negatively charged, hangs from a nylon thread so that it can turn freely. Omar rubs a test rod with a dry woollen cloth. He then holds the test rod near one end of the hanging rod, without touching it.

The table shows his results.

test rod	observation
polythene	the hanging rod turns away from the test rod
perspex	the hanging rod turns towards the test rod

(a) State the charge on the rubbed polythene test rod. Give a reason for your answer. [2]

(b) Omar says that the perspex rod must be positively charged.

Explain why this result alone does not show that the perspex rod is positively charged. [2]

(c) Describe how Omar could show that the perspex rod is positively charged. [2]

(d) State one variable Omar should keep the same each time he charges a test rod. [1]

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26 · Part 1

Paper 2 only

Electric charge

[6]

Priya rubs a rubber balloon on her dry hair. The balloon becomes negatively charged.

(a) Explain how the balloon becomes negatively charged. [2]

(b) State the charge on Priya's hair after rubbing. Explain your answer. [2]

(c) After rubbing, the individual hairs on Priya's head move apart from each other.

Explain why. [2]

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27 · Part 1 Paper 2 only Electric charge [6]

Fuel is pumped at high speed through a long hose into the fuel tank of an aircraft.

(a) Explain why refuelling an aircraft in this way can be dangerous. [3]

(b) Before refuelling starts, the aircraft is connected to the ground by a metal cable.

Explain how this reduces the danger. [2]

(c) Suggest one other situation in which a build-up of electrostatic charge could be dangerous. [1]

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28 · Part 1

Paper 2 only

Electric charge

[5]

In one type of inkjet printer, tiny drops of ink are given a negative charge. The drops then pass between two horizontal metal plates. The upper plate is positively charged and the lower plate is negatively charged.

(a) State the direction in which the charged drops are deflected. Explain your answer. [2]

(b) Explain how increasing the amount of charge on the plates changes where a drop lands on the paper. [2]

(c) One drop is not given any charge.

State what happens to this drop as it passes between the plates. [1]

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29 · Part 1

Paper 2 only

Electric charge

[6]

Photocopiers use electrostatic charge to make copies of documents.

(a) Describe how a photocopier uses electrostatic charge to produce a copy of a printed page. [5]

(b) All the particles of toner powder are given the same type of charge.

Suggest one reason why. [1]

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30 · Part 1

Paper 2 only

Electric charge

[6]

Ana wears shoes with rubber soles. She walks across a nylon carpet. When she touches a metal door handle, she feels a small electric shock.

(a) Explain why Ana feels an electric shock. [4]

(b) During the shock, a charge of 3.0×10^{-7} C flows in a time of 1.5×10^{-6} s.

Calculate the average current during the shock. [2]

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